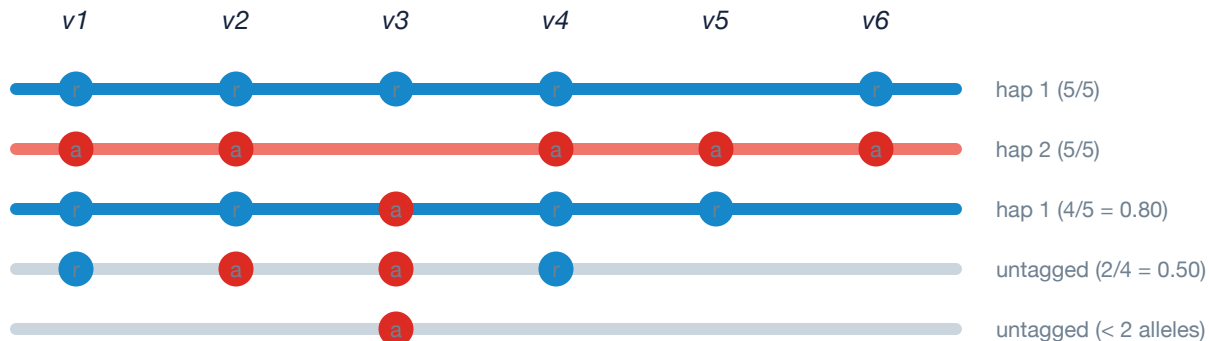


a Step 1: assign each read to a haplotype from the initial phase



A read is tagged when $\max(n_1, n_2)/(n_1 + n_2) > 0.65$ and it carries ≥ 2 informative alleles;
SNV and SV alleles count 1, indels 0.1, 5mC 0.

b Step 2: re-derive each variant's phase from tagged reads

For variant v_3 :

hap-1 reads with REF + hap-2 reads with ALT = $n_1 = 1$

hap-1 reads with ALT + hap-2 reads with REF = $n_2 = 2$

confidence $\rho = \max(n_1, n_2)/(n_1 + n_2) = 0.67 \leq 0.75$

→ v_3 left unphased (GT 0/1, no PS)

For variant v_1 :

$n_1 = 3, n_2 = 0, \rho = 1.00 > 0.75$

→ phase confirmed: REF on hap 1, ALT on hap 2 (GT 0|1)

Only variants supported by the two read populations keep a phase; assignments driven by a few chimeric or mis-mapped reads are removed. Thresholds: --readConfidence 0.65, --snpConfidence 0.75.